OOP Practical No: 04

Name: Tushar Balaso Solankar PRN: B26ET2021 Div: SY3 Class: C3

INPUT :

```
#include <iostream>
using namespace std;

class Complex
{
private:
  float real, imag;

public:

void read()
  {
    cout << "Enter real part: ";
    cin >> real;

    cout << "Enter imaginary part: ";
    cin >> imag;
  }

  // Addition
  Complex add(Complex c)
  {
    Complex result;
    result.real = real + c.real;
```

```
    result.imag = imag + c.imag;
    return result;
}

// Subtraction
Complex subtract(Complex c)
{
    Complex result;
    result.real = real - c.real;
    result.imag = imag - c.imag;
    return result;
}

// Multiplication
Complex multiply(Complex c)
{
    Complex result;
    result.real = (real * c.real) - (imag * c.imag);
    result.imag = (real * c.imag) + (imag * c.real);
    return result;
}

// Division
Complex divide(Complex c)
{
    Complex result;

    float denominator = (c.real * c.real) + (c.imag * c.imag);
```

```
    result.real = ((real * c.real) + (imag * c.imag)) / denominator;
    result.imag = ((imag * c.real) - (real * c.imag)) / denominator;

    return result;
}

// Complex conjugate
Complex conjugate()
{
    Complex result;
    result.real = real;
    result.imag = -imag;
    return result;
}

// Display complex number
void display()
{
    cout << real;

    if (imag >= 0)
        cout << " + " << imag << "i";
    else
        cout << " - " << -imag << "i";

    cout << endl;
}
```

```
};

int main()
{
    Complex c1, c2, result;

    cout << "Enter first complex number:" << endl;
    c1.read();

    cout << "\nEnter second complex number:" << endl;
    c2.read();

    cout << "\nFirst Complex Number: ";
    c1.display();

    cout << "Second Complex Number: ";
    c2.display();

    result = c1.add(c2);
    cout << "\nAddition: ";
    result.display();

    result = c1.subtract(c2);
    cout << "Subtraction: ";
    result.display();

    result = c1.multiply(c2);
    cout << "Multiplication: ";
```

```
    result.display();

    result = c1.divide(c2);
    cout << "Division: ";
    result.display();

    result = c1.conjugate();
    cout << "Conjugate of First Complex Number: ";
    result.display();

    return 0;
}
```

OUTPUT :

```
PS D:\oop\Practical-No-04-operations> .\code.exe
Enter first complex number:
Enter real part: 4
Enter imaginary part: 3

Enter second complex number:
Enter real part: 2
Enter imaginary part: 1

First Complex Number: 4 + 3i
Second Complex Number: 2 + 1i

Addition: 6 + 4i
Subtraction: 2 + 2i
Multiplication: 5 + 10i
Division: 2.2 + 0.4i
Conjugate of First Complex Number: 4 - 3i
PS D:\oop\Practical-No-04-operations>
PS D:\oop\Practical-No-04-operations>
```